

	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 25-10-2024	
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
August 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4715	1937	177
	4.2 < ws < 7.6	2626	3474	1565
	ws > 7.6	297	1654	8032
Number of points 24477 ; PC=66.3%; EBD=1.9%; POD1=61.7%; POD2=49.2%; POD3=82.2%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	7096	514	0
	4.2 < ws < 7.6	542	6158	378
	ws > 7.6	0	393	9396
Number of points 24477 ; PC=92.5%; EBD=0.0%; POD1=92.9%; POD2=87.2%; POD3=96.1%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
	ws < 12.0	20742	247	0
Model data	12.0 < ws < 18.0	159	2706	107
	ws > 18.0	0	76	440
Number of points 24477 ; PC=97.6%; EBD=0.0%; POD1=99.2%; POD2=89.3%; POD3=80.4%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

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wind direction degree		North	East	South	Observations	West
Model data	North	11576	624	45		140
	East	707	1109	452		5
	South	88	346	3330		267
	West	420	95	87		262
Number of points 19553 ; PC=83.2%; EBD=1.2%; POD1=90.5%; POD2=51.0%; POD3=85.1%; POD4=38.9%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	12632	179	0		30
	East	138	1910	59		0
	South	0	85	3848		28
	West	21	0	7		616
Number of points 19553 ; PC=97.2%; EBD=0.0%; POD1=98.8%; POD2=87.9%; POD3=98.3%; POD4=91.4%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	4247	1384	38
	7.1 < rh < 15.3	3343	5308	1244
	rh > 15.3	606	2071	5948
Number of points 24189 ; PC=64.1%; EBD=2.7%; POD1=51.8%; POD2=60.6%; POD3=82.3%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	7622	366	3
	7.1 < rh < 15.3	573	8169	232
	rh > 15.3	1	228	6995
Number of points 24189 ; PC=94.2%; EBD=0.0%; POD1=93.0%; POD2=93.2%; POD3=96.7%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	23600	44	0
	50.0 < rh < 70.0	30	356	4
	rh > 70.0	0	7	148
Number of points 24189 ; PC=99.6%; EBD=0.0%; POD1=99.9%; POD2=87.5%; POD3=97.4%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6183	1129	7
	1.9 < pwv < 3.5	798	4729	1100
	pwv > 3.5	8	699	7891
Number of points 22544 ; PC=83.4%; EBD=0.1%; POD1=88.5%; POD2=72.1%; POD3=87.7%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6877	159	0
	1.9 < pwv < 3.5	112	6315	92
	pwv > 3.5	0	83	8906
Number of points 22544 ; PC=98.0%; EBD=0.0%; POD1=98.4%; POD2=96.3%; POD3=99.0%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

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total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	2886	1723	554
Model data 0.6 < see < 0.8	1248	1550	1602
see > 0.8	200	690	1816
Number of points 12269 ; PC=51.0%; EBD=6.1%; POD1=66.6%; POD2=39.1%; POD3=45.7%			

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	4289	634	26
Model data 0.6 < see < 0.8	483	2974	359
see > 0.8	11	443	3856
Number of points 12269 ; PC=85.0%; EBD=0.3%; POD1=89.7%; POD2=73.4%; POD3=90.9%			

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	4171	0	390
Model data 0.6 < see < 0.8	0	3846	390
see > 0.8	163	117	3582
Number of points 12269 ; PC=94.5%; EBD=4.5%; POD1=96.2%; POD2=97.0%; POD3=82.1%			

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	4775	4	23
Model data 0.6 < see < 0.8	0	4027	22
see > 0.8	8	20	4218
Number of points 12269 ; PC=99.6%; EBD=0.2%; POD1=99.8%; POD2=99.4%; POD3=98.9%			

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2031	823	79
	$3.4 < \tau < 5.8$	665	1532	638
	$\tau > 5.8$	94	1237	3673
Number of points 10772 ; PC=67.2%; EBD=1.6%; POD1=72.8%; POD2=42.7%; POD3=83.7%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2618	208	1
	$3.4 < \tau < 5.8$	172	3090	264
	$\tau > 5.8$	0	294	4125
Number of points 10772 ; PC=91.3%; EBD=0.0%; POD1=93.8%; POD2=86.0%; POD3=94.0%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2540	137	336
	$3.4 < \tau < 5.8$	156	2812	336
	$\tau > 5.8$	94	643	4054
Number of points 10772 ; PC=87.3%; EBD=4.0%; POD1=91.0%; POD2=78.3%; POD3=85.8%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2787	1	14
	$3.4 < \tau < 5.8$	3	3550	14
	$\tau > 5.8$	0	41	4376
Number of points 10772 ; PC=99.5%; EBD=0.1%; POD1=99.9%; POD2=98.8%; POD3=99.4%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

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ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1804	1729
	0.6 < glf < 0.7	1080	1506
	glf > 0.7	328	563
Number of points 11053 ; PC=38.3%; EBD=15.3%; POD1=56.2%; POD2=39.7%; POD3=22.7%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2787	505
	0.6 < glf < 0.7	414	2744
	glf > 0.7	11	549
Number of points 11053 ; PC=81.0%; EBD=0.3%; POD1=86.8%; POD2=72.2%; POD3=84.6%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2842	437
	0.6 < glf < 0.7	42	3336
	glf > 0.7	328	25
Number of points 11053 ; PC=79.0%; EBD=16.5%; POD1=88.5%; POD2=87.8%; POD3=48.5%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3194	24
	0.6 < glf < 0.7	7	3759
	glf > 0.7	11	15
Number of points 11053 ; PC=99.2%; EBD=0.4%; POD1=99.4%; POD2=99.0%; POD3=98.4%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3801	845
	4.6 < ws < 7.4	3226	4152
	ws > 7.4	228	2174
Number of points 24002 ; PC=66.7%; EBD=1.6%; POD1=52.4%; POD2=57.9%; POD3=84.1%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	6501	524
	4.6 < ws < 7.4	753	6186
	ws > 7.4	1	461
Number of points 24002 ; PC=90.9%; EBD=0.0%; POD1=89.6%; POD2=86.3%; POD3=95.3%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	19558	306
	12.0 < ws < 18.0	214	2890
	ws > 18.0	0	103
Number of points 24002 ; PC=96.9%; EBD=0.0%; POD1=98.9%; POD2=87.6%; POD3=86.0%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	9305	588	16		210
	East	137	513	105		0
	South	24	56	2257		451
	West	2556	27	283		4474
Number of points 21002 ; PC=78.8%; EBD=0.3%; POD1=77.4%; POD2=43.3%; POD3=84.8%; POD4=87.1%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	11564	87	0		249
	East	91	1077	13		0
	South	0	18	2585		41
	West	367	2	63		4845
Number of points 21002 ; PC=95.6%; EBD=0.0%; POD1=96.2%; POD2=91.0%; POD3=97.1%; POD4=94.4%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4387	1313	8
	8.9 < rh < 17.9	3044	6195	1306
	rh > 17.9	257	1701	5716
Number of points 23927 ; PC=68.1%; EBD=1.1%; POD1=57.1%; POD2=67.3%; POD3=81.3%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	7279	355	0
	8.9 < rh < 17.9	409	8635	188
	rh > 17.9	0	219	6842
Number of points 23927 ; PC=95.1%; EBD=0.0%; POD1=94.7%; POD2=93.8%; POD3=97.3%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	23274	24	0
	50.0 < rh < 70.0	32	364	18
	rh > 70.0	0	9	206
Number of points 23927 ; PC=99.7%; EBD=0.0%; POD1=99.9%; POD2=91.7%; POD3=92.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5875	840	0
	2.1 < pwv < 3.7	1079	4551	1000
	pwv > 3.7	34	713	8417
Number of points 22509 ; PC=83.7%; EBD=0.2%; POD1=84.1%; POD2=74.6%; POD3=89.4%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6821	133	0
	2.1 < pwv < 3.7	167	5889	107
	pwv > 3.7	0	82	9310
Number of points 22509 ; PC=97.8%; EBD=0.0%; POD1=97.6%; POD2=96.5%; POD3=98.9%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4467	1885	169
	4.2 < ws < 7.6	2607	3398	1497
	ws > 7.6	369	1629	7818
Number of points 23839 ; PC=65.8%; EBD=2.3%; POD1=60.0%; POD2=49.2%; POD3=82.4%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	11355	668	82
	East	709	1046	650
	South	87	299	3013
	West	290	102	89

Number of points 19019 ; PC=82.0%; EBD=1.5%; POD1=91.3%; POD2=49.5%; POD3=78.6%; POD4=30.2%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3834	1285
	7.1 < rh < 15.3	3424	4811
	rh > 15.3	748	2301

Number of points 23525 ; PC=61.2%; EBD=3.4%; POD1=47.9%; POD2=57.3%; POD3=80.9%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6133	1255
	1.9 < pwv < 3.5	898	4260
	pwv > 3.5	19	907

Number of points 22365 ; PC=80.7%; EBD=0.4%; POD1=87.0%; POD2=66.3%; POD3=86.1%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3221	1812	756
	0.6 < see < 0.8	1520	2059	1909
	see > 0.8	375	1161	2715
Number of points		15528	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	4834	0	511
	0.6 < see < 0.8	0	4771	511
	see > 0.8	282	261	4869
Number of points		15528	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time ms	Observations			
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3201	1122	133
	$3.4 < \tau < 5.8$	801	1708	741
	$\tau > 5.8$	128	1527	4404
Number of points 13765 ; PC=67.7%; EBD=1.9%; POD1=77.5%; POD2=39.2%; POD3=83.4%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time ms	Observations			
		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3799	203	451
	$3.4 < \tau < 5.8$	203	3129	451
	$\tau > 5.8$	128	1025	4827
Number of points 13765 ; PC=85.4%; EBD=4.2%; POD1=92.0%; POD2=71.8%; POD3=84.3%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2070	2186
	0.6 < glf < 0.7	1721	2247
	glf > 0.7	329	591
Number of points 14152 ; PC=38.6%; EBD=14.1%; POD1=50.2%; POD2=44.7%; POD3=22.8%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3726	494
	0.6 < glf < 0.7	65	4501
	glf > 0.7	329	29
Number of points 14152 ; PC=80.5%; EBD=15.4%; POD1=90.4%; POD2=89.6%; POD3=48.4%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3811	840	200
	4.6 < ws < 7.4	3180	4131	1583
	ws > 7.4	252	2098	7785
Number of points 23880 ; PC=65.9%; EBD=1.9%; POD1=52.6%; POD2=58.4%; POD3=81.4%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	9124	643	19
	East	184	385	88
	South	26	57	2084
	West	2573	46	494

Number of points 20847 ; PC=77.0%; EBD=0.4%; POD1=76.6%; POD2=34.0%; POD3=77.6%; POD4=87.0%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3992	1015
	8.9 < rh < 17.9	3474	5632
	rh > 17.9	313	2397

Number of points 23793 ; PC=64.0%; EBD=1.4%; POD1=51.3%; POD2=62.3%; POD3=80.5%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5793	1028
	2.1 < pwv < 3.7	1564	4095
	pwv > 3.7	41	917

Number of points 22712 ; PC=78.7%; EBD=0.2%; POD1=78.3%; POD2=67.8%; POD3=86.1%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	4247	1989	233
	4.2 < ws < 7.6	2811	3243	1594
	ws > 7.6	411	1663	7715
Number of points 23906 ; PC=63.6%; EBD=2.7%; POD1=56.9%; POD2=47.0%; POD3=80.9%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	11251	751	89
	East	804	1005	666
	South	151	298	3013
	West	235	61	104

Number of points 19057 ; PC=81.1%; EBD=1.6%; POD1=90.4%; POD2=47.5%; POD3=77.8%; POD4=29.6%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3502	1429
	7.1 < rh < 15.3	3681	4849
	rh > 15.3	823	2186

Number of points 23592 ; PC=58.6%; EBD=3.8%; POD1=43.7%; POD2=57.3%; POD3=76.8%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6180	1498
	1.9 < pwv < 3.5	882	3798
	pwv > 3.5	57	1124

Number of points 22432 ; PC=77.0%; EBD=0.8%; POD1=86.8%; POD2=59.2%; POD3=82.0%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3187	1898	751
	0.6 < see < 0.8	1580	1941	1902
	see > 0.8	352	1190	2746
Number of points 15547 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	4816	0	518
	0.6 < see < 0.8	0	4829	518
	see > 0.8	303	200	4881
Number of points 15547 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3229	1074	72
	$3.4 < \tau < 5.8$	822	1696	860
	$\tau > 5.8$	131	1584	4350
Number of points 13818 ; PC=67.1%; EBD=1.5%; POD1=77.2%; POD2=39.0%; POD3=82.4%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3800	173	466
	$3.4 < \tau < 5.8$	251	3182	466
	$\tau > 5.8$	131	999	4816
Number of points 13818 ; PC=85.4%; EBD=4.3%; POD1=90.9%; POD2=73.1%; POD3=83.8%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1992	1997
	0.6 < glf < 0.7	1817	2449
	glf > 0.7	331	600
Number of points 14205 ; PC=38.6%; EBD=14.1%; POD1=48.1%; POD2=48.5%; POD3=20.8%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3724	397
	0.6 < glf < 0.7	85	4618
	glf > 0.7	331	31
Number of points 14205 ; PC=81.3%; EBD=15.1%; POD1=90.0%; POD2=91.5%; POD3=48.9%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3576	916
	4.6 < ws < 7.4	3235	4072
	ws > 7.4	383	2109
Number of points 23880 ; PC=64.4%; EBD=2.6%; POD1=49.7%; POD2=57.4%; POD3=80.6%			

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	8938	706	41
	East	143	337	104
	South	63	53	2031
	West	2824	41	509

Number of points 20869 ; PC=74.3%; EBD=0.7%; POD1=74.7%; POD2=29.6%; POD3=75.6%; POD4=82.7%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3617	1071
	8.9 < rh < 17.9	3759	5421
	rh > 17.9	383	2572

Number of points 23793 ; PC=60.3%; EBD=1.7%; POD1=46.6%; POD2=59.8%; POD3=76.1%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5615	1298
	2.1 < pwv < 3.7	1730	3759
	pwv > 3.7	72	946

Number of points 22759 ; PC=75.6%; EBD=0.9%; POD1=75.7%; POD2=62.6%; POD3=83.9%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration